

Falgun Batavia

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Summary

Penultimate Mechatronics and Computer Science student with hands-on experience across electrical systems, electronics design, and embedded platforms. Delivered complex electrical architectures for an electric vehicle and contributed to the design and electrical validation of solar cell arrays in a research-led engineering team. Highly motivated by hardware development and real-world engineering challenges.

Education

University of Sydney Jan 2023 - Nov 2027

- B.Eng (Mechatronics) (Hons) / B.Sc (Computer Science) - Weighted Average Mark (WAM): 81

- Recipient, Engineering Sydney Industry Placement Scholarship (ESIPS), 2026 semester 2 cohort

Tohoku University, Japan

Jul 2023

- Short-term exchange: Japanese Language and Culture Immersion

Relevant Experience

Engineering Intern, EurokaPower

Jan 2026 - Present

- Designed and tested PCB architectures for space-grade perovskite solar arrays.
- Developed Transfer Matrix Method (TMM) optical models in Python for multilayer/multi-junction solar cell structures across the 300 to 1200nm spectrum, analysing reflection, absorption, and parasitic optical losses, identifying performance limiting layers and increasing development efficiency.
- Contributed to documentation and Preliminary Design Review (PDR) and Critical Design Review (CDR) for 2027 space solar array launch.

Electrical Department Head, Sydney Motorsport (FSAE)

Jan 2025 - Dec 2025

- Led end-to-end design and integration of electrical systems for SM25, enabling the team's first fully functional electric vehicle, achieving compliance with FSAE standards under tight timelines.
- Conducted system-wide design reviews and managed a 15 member electrical team across High and Low voltage subsections under strict competition deadlines.

High Voltage Lead Engineer, Sydney Motorsport (FSAE)

Aug 2024 - Jan 2025

- Designed and validated a 470V, 6kWh accumulator, including cell selection and functional safety systems and circuits for the team's first ever EV, integrating this with motors and inverters.
- Led a team of 6 student engineers in the design and development of the aforementioned high voltage system.

Projects

Optical Modelling of Multi-Junction Solar Cells (Python):

- Programmed optical simulations for multilayer/multi-junction solar cell stacks in Python. Modelled parasitic loss, absorption/reflection, and External Quantum Efficiency (EQE).
- Correlated simulation results with lab measurements to optimise design layers for longevity and photocurrent generation of solar cells.

Beat-Sync VFX: Real-Time FPGA Audio-Visual System (Verilog, Altera FPGA):

- Created a DSP system for real-time beat detection from audio input to drive VGA display.
- Implemented VGA visual output with 5x5 convolution MAC array; achieved 1 pixel/clock throughput using pipelining and BRAM buffers.

Pulse Oximeter Printed Circuit Board (PCB) (Analog Optical Front-End):

- Built a two-layer analog PCB for SpO2/pulse sensing using red/IR LEDs and transimpedance amplifiers.
- Validated the signal via oscilloscope and LTspice; implemented active filtering for clean signal acquisition.

Battery Thermistor Acquisition PCB (STM32, C):

- Developed a CAN-enabled embedded board to monitor lithium-ion cell temperatures using an STM32 microcontroller.
- Used ADC sampling, fault-tolerant logic, and real-time CAN messaging for battery safety systems.

Skills

Hardware & System Design

PCB Design (Altium, KiCad), Analog/Digital Circuits, Schematic Capture, Harness Design (RapidHarness), Electrical System Design

Embedded & Firmware

FPGA, Verilog, SystemVerilog, STM32, Arduino, Raspberry Pi, ESP32, C, C++, Embedded C, Python, CAN Bus, SPI, I2C, UART

Simulation & Tools

Altium, Ansys HSFF, Fusion360, KiCad, LTspice, RapidHarness, MATLAB, Simulink, Python, Quartus, ModelSim, Git, Linux, SolidWorks

Soft Skills

Leadership, Project Management, Cross-Functional Collaboration, Interpersonal and Written Communication, Fast Learner, Research

References

Louise Gray - Previous Manager (Casual Academic Position)

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Daniel Jiang - Previous Tutor (Mechatronics 2, MTRX2700)

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